REDOX 3

18

Oxidising agents (oxidants) are substances that oxidise (take electrons away from) other substances. Common oxidising agents include potassium dichromate, concentrated tetraoxosulfuric acid, concentrated nitric acid and potassium permanganate.

1 Potassium dichromate has the formula $K_2Cr_2O_7$. Work out the oxidation state of chromium in this compound.

2 Tetraoxosulfuric acid (more commonly known as sulfuric acid) has the formula H_2SO_4. Work out the oxidation state of sulfur in this compound.

3 Potassium permanganate has the formula $KMnO_4$. Work out the oxidation state of manganese in this compound.

Potassium dichromate (VI) in an acidic solution oxidises alkanols (alcohols) to the corresponding alkanoic (carboxylic) acids; e.g. ethanol is oxidised to ethanoic (acetic) acid. (See figure 1.)

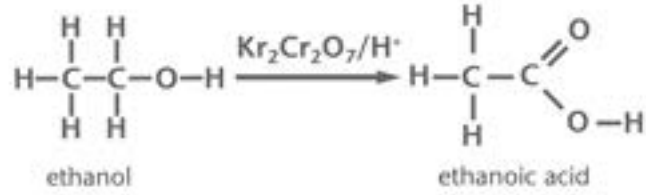


Figure 1

4 Give the molecular formula for ethanol.

5 Give the molecular formula for ethanoic acid

6 In this reaction, the orange dichromate (VI) ions are changed to green chromium (III) ions. Is chromium oxidised or reduced? Explain your answer.

Hydrogen bromide can be oxidised by a concentrated solution of sulfuric acid:

$$H_2SO_4\ (aq) + 2HBr\ (g) \rightarrow Br_2\ (aq) + SO_2\ (g) + 2H_2O\ (l)$$

7 Copy the equation above and calculate the oxidation states of the atoms present.

If copper metal is added to concentrated nitric acid, then water, copper nitrate and nitrogen dioxide can be formed:

$$Cu\ (s) + 4HNO_3\ (l) \rightarrow Cu(NO_3)_2\ (aq) + 2NO_2\ (g) + 2H_2O\ (l)$$

8 Copy the equation above and calculate the oxidation states of the atoms present.

Potassium permanganate is a strong oxidising agent. Solutions of potassium permanganate are useful for measuring concentrations of reducing agents. When potassium permanganate is dissolved in water it forms aqueous potassium (K^+) and manganate (MnO_4^-) ions. Potassium permanganate is frequently used in acidified solution and is purple in colour. When it is reduced, it forms the colourless manganese (Mn^{2+}) ion.

A typical reaction involving acidified permanganate solution is shown in part below:

$$MnO_4^-\ (aq) + aH^+\ (aq) + be^- \rightarrow Mn^{2+}\ (aq) + 4H_2O\ (l)$$

9 Calculate the number of hydrogen ions (a) present in the left side of the equation.

10 Calculate the number of electrons (b) present in the left side of the equation.

Acidified dichromate ($Cr_2O_7^{2-}$) ions behave in a similar manner to permanganate ions. Dichromate ions are reduced to chromium (Cr^{3+}) ions.

A typical reaction involving acidified dichromate solution is shown in part below:

$$Cr_2O_7^{2-}\ (aq) + cH+\ (aq) + de^- \rightarrow 2Cr^{3+}\ (aq) + 7H_2O\ (l)$$

11 Calculate the number of hydrogen ions (c) present in the left side of the equation.

12 Calculate the number of electrons (d) present in the left side of the equation

13 Write a balanced equation for each of these redox reactions. Show the oxidation states of the atoms present.

a One process that occurs in a lead-acid accumulator (car battery) is lead metal and lead (IV) oxide in sulfuric acid producing lead (II) sulfate and water.

b When potassium dichromate solution is added to hydrochloric acid, chlorine gas, water, chromium (III) chloride and potassium chloride are formed.

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Galvanic cells Redox reactions involve the transfer of electrons, so these reactions can be used to generate an electric current. Electrochemical cells that generate electricity are called Galvanic cells, named in honour of Luigi Galvani (1737–1798). Galvani investigated the effect of electricity on muscles. In turn, his research led Alessandro Volta (1745–1827) to develop (around 1800) the first continuous chemical source of electricity. Volta placed discs of silver, cardboard soaked in brine (sodium chloride solution) and zinc in repeating layers. One end had a silver plate on the outside, the other end had zinc. When he joined the silver and the zinc with a wire, electricity flowed continuously through the wire. Figure 2 shows one way of looking at this experiment in a school laboratory. If separate zinc and silver electrodes are placed in an electrolyte such as sodium chloride solution, reactions occur and a current flows. (An **electrolyte** is a liquid that allows a current to flow because there are mobile ions present.) The wire connecting the electrodes (via a voltmeter) allows electrons to flow. The following observations can also be made:

- The zinc electrode is negative
- The silver electrode is positive
- The electrodes change in appearance.

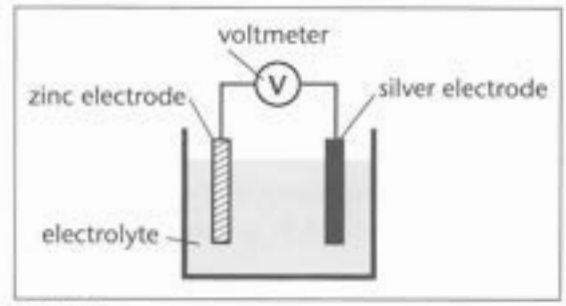


Figure 2

14 What evidence is there to show that electrons flow from the zinc electrode to the silver electrode?

15 Redraw figure 2, showing the charges on both electrodes and the direction of current from zinc electrode to silver electrode in the external circuit.

The simple cell shown in figure 2 generates current for only a short time because an imbalance of charge builds up. In order to get a clearer idea of the reactions involved, a more complex cell has to be set up. Figure 3 shows a zinc electrode in a solution of a zinc salt and a silver electrode in a solution of a silver salt. A salt bridge (e.g. KNO_3 or NH_4Cl, gel or solution) allows ions to move (and hence completes the circuit) but minimises the contamination of the two solutions. Ions need to move to balance the flow of electrons.

The following observations can also be made:

- The zinc electrode loses mass
- The silver electrode gains mass.

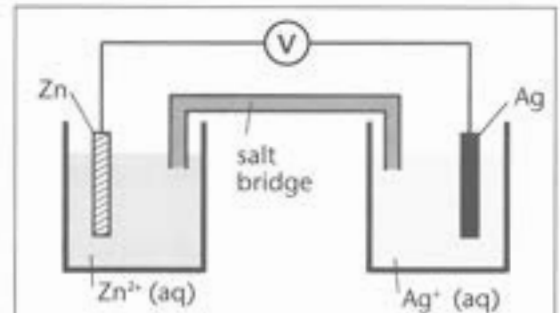


Figure 3

16 Redraw figure 3, showing the charges on both electrodes and the direction of current from zinc electrode to silver electrode.

17 Outline the purpose of the salt bridge.

18 Name a suitable substance for use as a salt bridge.

19 At the zinc electrode, zinc metal is forming zinc (Zn^{2+}) ions. Write an equation for this reaction. State whether it is oxidation or reduction.

20 At the silver electrode, silver (Ag^+) ions are forming silver metal. Write an equation for this reaction. State whether it is oxidation or reduction.

21 The answers to Q19 and Q20 are *half-equations*. Adjust your equations for Q19 and Q20 so that they have the same number of electrons present. Add the two half-equations to show the overall reaction.

22 Reduction takes place at the *cathode* and oxidation takes place at the *anode*.

a Which electrode is the cathode?

b Which electrode is the anode?

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